

CASH FLOW STATEMENT PREPARATION MANNIX INTL. CORP. - PART II

Chapters 8 and 10



**Universit 
Bocconi**
GRADUATE SCHOOL

20221 - FINANCIAL STATEMENTS ANALYSIS (ADVANCED COURSE)

Given data are:

- ☐ a “reformulated” balance sheet (managerial format)
- ☐ a “reformulated” income statement (Net Income + O.C.I.)
- ☐ some additional information (from notes to FS)

To practice we will see 3 cash flow statements:

1. “standard” cash flow statement (including in the Cash Flow from Operating Activities the effects of all the income statement items using the indirect method)
2. “reformulated” cash flow statement (change in cash & cash equivalents with OCF and OFCF)
3. “reformulated” cash flow statement (change in net financial position with OCF and OFCF)

“Standard” CFS (change in C&CE)

Cash Flow from Operating Activities
Cash Flow from Investing Activities
Cash Flow from Financing Activities
Net Change in Cash and Cash Equivalents

“Reformulated” CFS (OCF w/change in C&CE)

Operating Cash Flow - CAPEX
Operating Free Cash Flow
Other Cash Flows
Net Change in Cash and Cash Equivalents

Reformulation without changing the bottom line

“Reformulated” CFS (OCF w/change in Net Financial Position)

Operating Cash Flow - CAPEX
Operating Free Cash Flow
Other Cash Flows
Net Change in Net Financial Position

Change of the bottom line (excluding the cash flows that do not affect the size of the NFP)



Net Profit (Loss)	8.800
Depreciation and Amortization	12.000
Impairment of Tangible Assets	4.000
Impairment reversal (financial income)	(3.000)
(Gains)Losses on Sale of PPE	(7.000)
(Gains)Losses on Sale of Long-term Financial Assets	(28.000)
"Monetary" Income	(13.200)
-/+ increase / decrease of Accounts Receivable, net	19.000
-/+ increase / decrease of Inventory	24.000
+/- increase / decrease of Accounts Payable	(15.000)
+/- increase / decrease of Provisions for Product Warranty Costs	2.000
+/- increase / decrease of Provision for Litigation Risks	11.000
+/- increase / decrease of Tax Payables (for Income Taxes)	3.900
-/+ increase / decrease of Net Working Capital	44.900

Cash Flow from Operating Activities	31.700
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The total cash flow was positive (+56.200) but remember what we know now, mainly:

- **Negative “monetary” net income (-13.200)**
- **Strong decrease of Working Capital (+44.900)**
- **A nonrecurring disposal of Financial Assets (+53.000)**

Purchase of Property Plant and Equipment	(20.000)
Change in A/P for purchase of PPE	2.000
Investment in Property Plant and Equipment	(18.000)
Sale of Property Plant and Equipment	5.000
Gains(Losses) on Sale of PPE	7.000
Change in Receivable for sale of PPE	0
Divestment of Property Plant and Equipment	12.000
Purchase of Long-term Financial Assets	(4.000)
Change in Payables for purchase of long-term financial assets	0
Investment in Long-term Financial Assets	(4.000)
Sale of Long-term Financial Assets	25.000
Gains(Losses) on Sale of Long-term Financial Assets	28.000
Change in Receivables for sale of Long-term Financial Assets	0
Divestment of Long-term Financial Assets	53.000
Change in Marketable Securities	(6.000)
Change in AOCI (revaluation) for Marketable Securities	1.500
Net Investment in Marketable Securities	(4.500)

Cash Flow from Investing Activities	38.500
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Payment of the current portion of Long-term Financial Loans	(15.000)
Proceeds from Long-term Financial Loans	9.000
Net change in Short-term Bank Loans	(20.000)
Issuance of Share Capital (cash contributions)	25.000
Change in Subscriptions Receivable	(8.000)
Proceeds from issuance of Share Capital	17.000
Dividends Paid	(5.000)

Cash Flow from Financing Activities	(14.000)
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Net Change in Cash & Cash Equivalents	56.200
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Cash & Cash Equivalents at beginning of year	2.000
Net Increase (decrease) in Cash & Cash Equivalents	56.200
Cash & Cash Equivalents at end of year	58.200

2021 - FINANCIAL STATEMENTS ANALYSIS (ADVANCED COURSE)



The get the operating Cash Flow (with the indirect method) we must:

1. replace the starting point (Operating Income instead of Net Income) and
2. apply the (usual) 3 adjustments for items that relate to the Operating revenues and costs.

1. non monetary

1. non monetary

2. Unusual item

start →

From the Income Statement:

Net Sales revenues	185.000
Consumptions of Materials	(85.000)
Purchase of Services	(35.000)
Payroll Costs	(40.000)
Impairment of Tangible Assets	(4.000)
Allowance for Doubtful Receivables	(2.500)
Depreciation and Amortization	(12.000)
Provisions for Litigations Risks	(15.000)
Product Warranty Costs (net of Provisions Utilization)	(1.000)
Operating Income	(9.500)

From the Cash Flow from Operating Activities:

3. NOWC	-/+ increase / decrease of Accounts Receivable, net	19.000
3. NOWC	-/+ increase / decrease of Inventory	24.000
3. NOWC	+/- increase / decrease of Accounts Payable	(15.000)
3. NOWC	+/- increase / decrease of Provisions for Product Warranty Costs	2.000
Unusual	+/- increase / decrease of Provision for Litigation Risks	11.000
Before taxes	+/- increase / decrease of Tax Payables (for Income Taxes)	3.900
	-/+ increase / decrease of Net Working Capital	44.900

Additional note on Adjustment 2 (classification to other sections of the CFS); a comparison with what we did for the “standard” CFS:

- **Gains/losses from asset disposal** are classified in the I/S as unusual items, so below the Operating Income; starting from this metric we do not need any adjustment;
- **Provisions for litigation risks**: since the litigation settlement expense (and the related utilization of the provision liability) are classified in the I/S as unusual items, we must do the same for the provision expense, excluding this item from the calculation of the Operating Cash Flow



The additional adjustment of -500 must be recorded for two reasons:

1) The write-off (net of allowance utilization) is classified as Unusual Item (-500). It constitutes a “unusual” decrease of A/R (uncollectible A/R recorded in previous years).

2) We have a decrease of A/R because of losses and not because of collections.

Operating Income	(9.500)	
Depreciation and Amortization	12.000	
Impairment of Tangible Assets	4.000	
Provisions for Litigations Risks	15.000	
Net Operating Working Capital Flow	21.500	It is the EBITDA
-/+ increase / decrease of Accounts Receivable, net	19.000	
-/+ increase / decrease of A/R write-off net of allowance utilization	(500)	
-/+ increase / decrease of Inventory	24.000	
+/- increase / decrease of Accounts Payable	(15.000)	
+/- increase / decrease of Provisions for Product Warranty Costs	2.000	
-/+ increase / decrease of Net Operating Working Capital	29.500	
Operating Cash Flow (A)	51.000	

From the Balance Sheet:

	2019	2020	Change
Accounts receivable	65.000	45.000	(20.000)
Allowance for Doubtful Receivables	(3.000)	(2.000)	1.000
Accounts Receivable, net	62.000	43.000	(19.000)

From the Income Statement:

Earning before Unusual Items and Taxes	(13.800)
Write-off of receivables (recorded in past years)	(4.000)
Utilization: Allowance for Doubtful Receivables	3.500
Gains(Losses) from Assets Disposals	35.000
Litigation Settlement Expense	(10.000)
Utilization: Provisions for Litigation Risks	4.000
Income before Taxes	14.700



Let's select now the CAPEX from the Cash Flow form Investing Activities:



Purchase of Property Plant and Equipment	(20.000)	
Change in A/P for purchase of PPE	2.000	
Investment in Property Plant and Equipment		(18.000)
Sale of Property Plant and Equipment	5.000	
Gains(Losses) on Sale of PPE	7.000	
Change in Receivable for sale of PPE	0	
Divestment of Property Plant and Equipment		12.000
Purchase of Long-term Financial Assets	(4.000)	
Change in Payables for purchase of long-term financial assets	0	
Investment in Long-term Financial Assets		(4.000)
Sale of Long-term Financial Assets	25.000	
Gains(Losses) on Sale of Long-term Financial Assets	28.000	
Change in Receivables for sale of Long-term Financial Assets	0	
Divestment of Long-term Financial Assets		53.000
Change in Marketable Securities	(6.000)	
Change in AOCI (revaluation) for Marketable Securities	1.500	
Net Investment in Marketable Securities		(4.500)

Cash Flow from Investing Activities

38.500



This EBITDA is calculated:

- **before non-monetary items** (impairment and depreciation / amortization)
- **before any unusual items** (ex.. provision expense for litigation risks), if they have not already been reclassified....

Operating Income	(9.500)
Depreciation and Amortization	12.000
Impairment of Tangible Assets	4.000
Provisions for Litigations Risks	15.000
EBITDA	21.500

-/+ increase / decrease of Accounts Receivable, net	19.000
-/+ increase / decrease of A/R write-off net of allowance utilization	(500)
-/+ increase / decrease of Inventory	24.000
+/- increase / decrease of Accounts Payable	(15.000)
+/- increase / decrease of Provisions for Product Warranty Costs	2.000
-/+ increase / decrease of Net Operating Working Capital	29.500

Operating Cash Flow (A) 51.000

Purchase of Property Plant and Equipment	(20.000)
Change in A/P for purchase of PPE	2.000
Investment in Property Plant and Equipment	(18.000)
Sale of Property Plant and Equipment	5.000
Gains(Losses) on Sale of PPE	7.000
Change in Receivable for sale of PPE	0
Divestment of Property Plant and Equipment	12.000

Capital Expenditure CAPEX (B) (6.000)

Operating Free Cash Flow (A+B) 45.000



Cash flow for *Non-Operating Income* items

Financial Income	3.500
Impairment reversal (financial income)	(3.000)
Proceeds from Financial Income	500
Payment of Financial Expenses	(7.800)
Payment of Litigation Settlement	(10.000)
Income Taxes	(5.900)
Change in Tax Payables (for Income Taxes)	3.900
Payment of Income Taxes	(2.000)

Cash flow from *Non-Operating Investments* (financial assets)

Purchase of Long-term Financial Assets	(4.000)
Change in Payables for purchase of long-term financial assets	0
Investment in Long-term Financial Assets	(4.000)
Sale of Long-term Financial Assets	25.000
Gains(Losses) on Sale of Long-term Financial Assets	28.000
Change in Receivables for sale of Long-term Financial Assets	0
Divestment of Long-term Financial Assets	53.000
Change in Marketable Securities	(6.000)
Change in AOCI (revaluation) for Marketable Securities	1.500
Net Investment in Marketable Securities	(4.500)

Cash flow from *Financing Activities*

Payment of the current portion of Long-term Financial Loans	(15.000)
Proceeds from Long-term Financial Loans	9.000
Net change in Short-term Bank Loans	(20.000)
Issuance of Share Capital (cash contributions)	25.000
Change in Subscriptions Receivable	(8.000)
Proceeds from issuance of Share Capital	17.000
Dividends Paid	(5.000)

Other Cash Flows **11.200**

Net Change in Cash & Cash Equivalents **56.200**

The same results we got with the “standard” CFS

Cash & Cash Equivalents at beginning of year	2.000
Net Increase (decrease) in Cash & Cash Equivalents	56.200
Cash & Cash Equivalents at end of year	58.200

Income Statement		"Monetary" Income	With EBITDA	Notes
Net Sales revenues	185.000	185.000	185.000	
Consumptions of Materials	-85.000	-85.000	-85.000	
Purchase of Services	-35.000	-35.000	-35.000	
Payroll Costs	-40.000	-40.000	-40.000	
Impairment of Tangible Assets	-4.000	NO	NO	Non monetary
Allowance for Doubtful Receivables	-2.500	-2.500	-2.500	
Depreciation and Amortization	-12.000	NO	NO	Non monetary
Provisions for Litigations Risks	-15.000	-15.000	Unusual	Reclassification to Unusual items
Product Warranty Costs (net of Provisions Utilization)	-1.000	-1.000	-1.000	
Operating Income	-9.500	6.500	21.500	To be matched w/Δ NOWC
Financial Income	3.500	500	500	Inflow
Financial Expenses	-7.800	-7.800	-7.800	Outflow
Earning before Unusual Items and Taxes	-13.800	-800	14.200	
Write-off of receivables (recorded in past years)	-4.000	-4.000	-4.000	Adj. of Δ A/R
Utilization: Allowance for Doubtful Receivables	3.500	3.500	3.500	
Gains(Losses) from Assets Disposals	35.000	NO	NO	Reclassification to Investing Activities
Litigation Settlement Expense	-10.000	-10.000	-10.000	To be matched w/Δ Provision liability (+11,000):
Provisions for Litigations Risks			-15.000	-21,000+11,000=-10,000
Utilization: Provisions for Litigation Risks	4.000	4.000	4.000	
Income before Taxes	14.700	-7.300	-7.300	
Income Taxes	-5.900	-5.900	-5.900	To be matched w/Δ Tax payable (+3,900):
				-5,900+3,900=-2,000
PROFIT (LOSS) FOR THE YEAR	8.800	-13.200	-13.200	
<u>Other Comprehensive Income:</u>				
Revaluation of Available for Sale marketable securities	1.500	NO	NO	Non cash (revaluation)
TOTAL COMPREHENSIVE INCOME FOR THE YEAR	10.300	-13.200	-13.200	



From the Balance Sheet:

<i>Net Financial Position</i>	2019	2020	Change
Short-term Bank Loans	32.000	12.000	-20.000
Current Portion of Long-term Financial Loans	15.000	18.000	3.000
Long-term Financial Loans	65.000	61.000	-4.000
Cash and Bank Accounts	-2.000	-58.200	-56.200
Short-term Marketable Securities (Available for Sale)	-12.000	-18.000	-6.000
	98.000	14.800	-83.200

In the Reformulated Statement of change in Net Financial Position the “*Other Cash Flows*” exclude the cash flows related to those balance sheet accounts that are included into the Net Financial Position:

- short-term and long-term financial liabilities
- short-term financial assets

My rule: “if something is inside the bottom line it can’t stay above the bottom line”

Reformulated Cash Flow Statement (change in Cash and OFCF)

Operating Free Cash Flow (A+B)	45.000
Financial Income	3.500
Impairment reversal (financial income)	(3.000)
Proceeds from Financial Income	500
Payment of Financial Expenses	(7.800)
Payment of Litigation Settlement	(10.000)
Income Taxes	(5.900)
Change in Tax Payables (for Income Taxes)	3.900
Payment of Income Taxes	(2.000)
Purchase of Long-term Financial Assets	(4.000)
Change in Payables for purchase of long-term financial assets	0
Investment in Long-term Financial Assets	(4.000)
Sale of Long-term Financial Assets	25.000
Gains(Losses) on Sale of Long-term Financial Assets	28.000
Change in Receivables for sale of Long-term Financial Assets	0
Divestment of Long-term Financial Assets	53.000
Change in Marketable Securities	(6.000)
Change in AOCI (revaluation) for Marketable Securities	1.500
Net Investment in Marketable Securities	(4.500)
Payment of the current portion of Long-term Financial Loans	(15.000)
Proceeds from Long-term Financial Loans	9.000
Net change in Short-term Bank Loans	(20.000)
Issuance of Share Capital (cash contributions)	25.000
Change in Subscriptions Receivable	(8.000)
Proceeds from issuance of Share Capital	17.000
Dividends Paid	(5.000)
Other Cash Flows	11.200
Net Change in Cash & Cash Equivalents	56.200

<i>Cash & Cash Equivalents at beginning of year</i>	<i>2.000</i>
<i>Net Increase (decrease) in Cash & Cash Equivalents</i>	<i>56.200</i>
<i>Cash & Cash Equivalents at end of year</i>	<i>58.200</i>

Reformulated Cash Flow Statement (change in NFP and OFCF)

Operating Free Cash Flow (A+B)	45.000
Financial Income	3.500
Impairment reversal (financial income)	(3.000)
Proceeds from Financial Income	500
Payment of Financial Expenses	(7.800)
Payment of Litigation Settlement	(10.000)
Income Taxes	(5.900)
Change in Tax Payables (for Income Taxes)	3.900
Payment of Income Taxes	(2.000)
Purchase of Long-term Financial Assets	(4.000)
Change in Payables for purchase of long-term financial assets	0
Investment in Long-term Financial Assets	(4.000)
Sale of Long-term Financial Assets	25.000
Gains(Losses) on Sale of Long-term Financial Assets	28.000
Change in Receivables for sale of Long-term Financial Assets	0
Divestment of Long-term Financial Assets	53.000
Issuance of Share Capital (cash contributions)	25.000
Change in Subscriptions Receivable	(8.000)
Proceeds from issuance of Share Capital	17.000
Dividends Paid	(5.000)
Other Cash Flows	41.700
Net Change in Net Financial Position	86.700

Reformulated Cash Flow Statement (change in Cash and OFCF)

Reformulated Cash Flow Statement (change in NFP and OFCF)

Operating Free Cash Flow (A+B) 45.000

Operating Free Cash Flow (A+B) 45.000

We need to reconcile the overall change in Net Financial Position:

Net Financial Position

Short-term Bank Loans
Current Portion of Long-term Financial Loans
Long-term Financial Loans
Cash and Bank Accounts
Short-term Marketable Securities (Available for Sale)

2019	2020	Change
32.000	12.000	-20.000
15.000	18.000	3.000
65.000	61.000	-4.000
-2.000	-58.200	-56.200
-12.000	-18.000	-6.000
98.000	14.800	-83.200

- The cash flow is +86.700 (decrease of the NFP by -86.700)
- The overall change of the NFP is a decrease of -83.200
- The difference is 3.500 (a net increase of the NFP): why?

Financial Income
Impairment reversal (fi
Proceeds from Financial
Payment of Financial Ex
Payment of Litigation Se
Income Taxes
Change in Tax Payable
Payment of Income Tax

Purchase of Long-term
Change in Payables fo
Investment in Long-term
Sale of Long-term Fin
Gains(Losses) on Sale
Change in Receivables
Divestment of Long-term
Change in Marketable
Change in AOCI (reval
Net Investment in Marke

Payment of the current p
Proceeds from Long-term Financial Loans 9.000
Net change in Short-term Bank Loans (20.000)
Issuance of Share Capital (cash contributions) 25.000
Change in Subscriptions Receivable (8.000)
Proceeds from issuance of Share Capital 17.000
Dividends Paid (5.000)

Other Cash Flows 11.200

Net Change in Cash & Cash Equivalents 56.200

Cash & Cash Equivalents at beginning of year 2.000
Net Increase (decrease) in Cash & Cash Equivalents 56.200
Cash & Cash Equivalents at end of year 58.200

3.500
(3.000)
500
(7.800)
(10.000)
(5.900)
3.900
(2.000)
(4.000)
0
(4.000)
25.000
28.000
0
53.000
25.000
(3.000)
17.000
(5.000)

Other Cash Flows 41.700

Net Change in Net Financial Position 86.700

Reformulated Cash Flow Statement (change in Cash and OCF)

Operating Free Cash Flow (A+B)	45.000
Financial Income	3.500
Impairment reversal (financial income)	(3.000)
Proceeds from Financial Income	500
Payment of Financial Expenses	(7.800)
Payment of Litigation Settlement	(10.000)
Income Taxes	(5.900)
Change in Tax Payables (for Income Taxes)	3.900
Payment of Income Taxes	(2.000)
Purchase of Long-term Financial Assets	(4.000)
Change in Payables for purchase of long-term financial assets	0
Investment in Long-term Financial Assets	(4.000)
Sale of Long-term Financial Assets	25.000
Gains(Losses) on Sale of Long-term Financial Assets	28.000
Change in Receivables for sale of Long-term Financial Assets	0
Divestment of Long-term Financial Assets	53.000
Change in Marketable Securities	(6.000)
Change in AOCI (revaluation) for Marketable Securities	1.500
Net Investment in Marketable Securities	(4.500)
Payment of the current portion of Long-term Financial Loans	(15.000)
Proceeds from Long-term Financial Loans	9.000
Net change in Short-term Bank Loans	(20.000)
Issuance of Share Capital (cash contributions)	25.000
Change in Subscriptions Receivable	(8.000)
Proceeds from issuance of Share Capital	17.000
Dividends Paid	(5.000)
Other Cash Flows	11.200
Net Change in Cash & Cash Equivalents	56.200

Cash & Cash Equivalents at beginning of year 2.000
Net Increase (decrease) in Cash & Cash Equivalents 56.200
Cash & Cash Equivalents at end of year 58.200

Reformulated Cash Flow Statement (change in NFP and OCF)

Operating Free Cash Flow (A+B)	45.000
Financial Income	3.500
Impairment reversal (financial income)	(3.000)
Proceeds from Financial Income	500
Payment of Financial Expenses	(7.800)
Payment of Litigation Settlement	(10.000)
Income Taxes	(5.900)
Change in Tax Payables (for Income Taxes)	3.900
Payment of Income Taxes	(2.000)
Purchase of Long-term Financial Assets	(4.000)
Change in Payables for purchase of long-term financial assets	0
Investment in Long-term Financial Assets	(4.000)
Sale of Long-term Financial Assets	25.000
Gains(Losses) on Sale of Long-term Financial Assets	28.000
Change in Receivables for sale of Long-term Financial Assets	0
Divestment of Long-term Financial Assets	53.000
Issuance of Share Capital (cash contributions)	25.000
Change in Subscriptions Receivable	(8.000)
Proceeds from issuance of Share Capital	17.000
Dividends Paid	(5.000)
Other Cash Flows	41.700
Net Change in Net Financial Position	86.700
Net Financial Position at beginning of year	98.000
+/-negative/positive Cash Flow	-86.700
+ Increase in Long-term Financial Loans for Lease of PPE	5.000
+ Revaluation of Available for Sale marketable securities	-1.500
Net Financial Position at end of year	14.800

<i>Net Financial Position</i>	2019	2020	Change		Cash	Reclass.	Non Cash	Change
Short-term Bank Loans	32.000	12.000	-20.000		-20.000			-20.000
Current Portion of Long-term Financial Loans	15.000	18.000	3.000		-15.000	18.000		3.000
Long-term Financial Loans	65.000	61.000	-4.000		9.000	-18.000	5.000	-4.000
Cash and Bank Accounts	-2.000	-58.200	-56.200		-56.200			-56.200
Short-term Marketable Securities	-12.000	-18.000	-6.000		-4.500		-1.500	-6.000
NET FINANCIAL POSITION	98.000	14.800	-83.200		-86.700	0	3.500	-83.200

Net Financial Position at beginning of year	98.000
+/-negative/positive Cash Flow	-86.700
+ Increase in Long-term Financial Loans for Lease of PPE	5.000
+ Revaluation of Available for Sale marketable securities	-1.500
Net Financial Position at end of year	14.800

CAN WE IDENTIFY THE CASH AND NON-CASH CHANGES OF THE NFP?

- For the “*Cash*” change of the NFP you may refer to the Cash Flow Statement where the bottom line is the *change in cash & cash equivalents*.
- The column “*Reclass.*” shows that at the end of the year a portion of the LT financial loans was reclassified into the “*current portion*” item.
- For the “*Non Cash*” change of the NFP you may refer to the reconciliation of the NFP:
 - Increase of LT financial loans because of the lease contracts (+5.000): increase of the NFP
 - Increase of marketable securities because of a revaluation (+1.500): decrease of the NFP

